

University Hostel Management Sytem

TA: Akram Tarek

Team ID: 23

Team Members:

Student 1

Name: أحمد هاني السيد أحمد

Seat Number: 20241700077 Academic Level: 2nd Department: No Departments

Student 2

Name: حسن عصام محمد محمد

Seat Number: 20241700259 Academic Level: 2nd Department: No Department

Student 3

Name: أحمد وائل محمد علي

Seat Number: 20241700083 Academic Level: 2nd Department: No Department

Student 4

Name: حسن علاء مختار حسن

Seat Number: 20241700260 Academic Level: 2nd Department: No Department

Student 5

Name: مايكل سامح فرج رزق

Seat Number: 20241700681 Academic Level: 2nd Department: No Department

Student 6

Name: حسن محمد جمال سيد

Seat Number: 20241700261 Academic Level: 2nd Department: No Department

Student 7

Name: حازم احمد محمد احمد

Seat Number: 20241700236 Academic Level: 2nd Department: No Department

Student 8

Name: أحمد هاني ز غلول محمد

Seat Number: 20241700080 Academic Level: 2nd Department: No Department

SQL Script:

Creation:

```
-- create table hostel
CREATE table hostel (
hostel_id NUMBER PRIMARY key,
hostel_name varchar(40) not null,
hostel_capacity NUMBER not null
);

-- create table room
CREATE table room (
room_id NUMBER PRIMARY key,
floor_no NUMBER not null,
room_capacity NUMBER not null,
status varchar(40) check (status in ('Vacant', 'Occupied', 'Under Maintenance')),
hostel_id NUMBER REFERENCES hostel(hostel_id)
);

-- create table student
CREATE table student (
s_id NUMBER PRIMARY key,
first_name varchar(40) not null,
last_name varchar(40) not null,
gender char(1) check (gender in ('F','M')),
DOB date not null,
st_date date not null,
hostel_id NUMBER REFERENCES hostel(hostel_id),
room_id NUMBER REFERENCES room (room_id)
);

-- create student_phone
CREATE table student_phone (
s_id NUMBER REFERENCES student(s_id),
phone varchar(15),
PRIMARY key(s_id, phone)
);

--staff table
CREATE table staff (
staff_id NUMBER PRIMARY key,
name varchar(40) not null,
ROLE varchar(40) not null,
hostel_id NUMBER REFERENCES hostel(hostel_id),
staff_supervisor_id NUMBER REFERENCES staff(staff_id), -- when we add first staff
```

```
we should put a staff_sup with null
salary NUMBER not null
);
```

```
-- staff_phone table
CREATE table staff_phone (
staff_id NUMBER REFERENCES staff(staff_id),
phone varchar(15)
);
```

```
-- fee table
CREATE table fee (
payment NUMBER PRIMARY key,
ammount NUMBER not null,
pay_method varchar(20) not null,
pay_date date not null,
s_id NUMBER REFERENCES student(s_id)
);
```

Insert:

```
-- HOSTEL
INSERT INTO HOSTEL VALUES (1, 'Al-Noor Hostel', 120);
INSERT INTO HOSTEL VALUES (2, 'Al-Fajr Hostel', 100);
INSERT INTO HOSTEL VALUES (3, 'Al-Ihsan Hostel', 150);

-- ROOM
INSERT INTO ROOM VALUES (101, 1, 4, 'Occupied', 1);
INSERT INTO ROOM VALUES (102, 1, 2, 'Vacant', 1);
INSERT INTO ROOM VALUES (201, 2, 4, 'Occupied', 2);

-- STUDENT
INSERT INTO STUDENT VALUES (1001, 'Hazem', 'Ahmed', 'M', DATE '2003-08-15',
DATE '2024-09-01', 1, 101);
INSERT INTO STUDENT VALUES (1002, 'Sara', 'Hassan', 'F', DATE '2004-03-10',
DATE '2024-09-05', 1, 101);
INSERT INTO STUDENT VALUES (1003, 'Hassan', 'Mohamed', 'M', DATE '2003-11-22',
DATE '2024-09-05', 2, 201);

-- STUDENT_PHONE
INSERT INTO STUDENT_PHONE VALUES (1001, '01156867798');
INSERT INTO STUDENT_PHONE VALUES (1001, '01227976900');
INSERT INTO STUDENT_PHONE VALUES (1002, '01120568727');

-- STAFF (Note: supervisor = 5003 for demo - see below)
INSERT INTO STAFF VALUES (5003, 'Tariq Sami', 'Supervisor', 1, NULL, 10000);
INSERT INTO STAFF VALUES (5001, 'Ahmed Mohamed', 'Warden', 1, 5003, 8000);
```

INSERT INTO STAFF VALUES (5002, 'Layla Nasser', 'Cleaner', 1, 5003, 3000);

-- STAFF_PHONE

INSERT INTO STAFF_PHONE VALUES (5001, '01152198657');

INSERT INTO STAFF_PHONE VALUES (5002, '01152197657');

INSERT INTO STAFF_PHONE VALUES (5003, '01152100657');

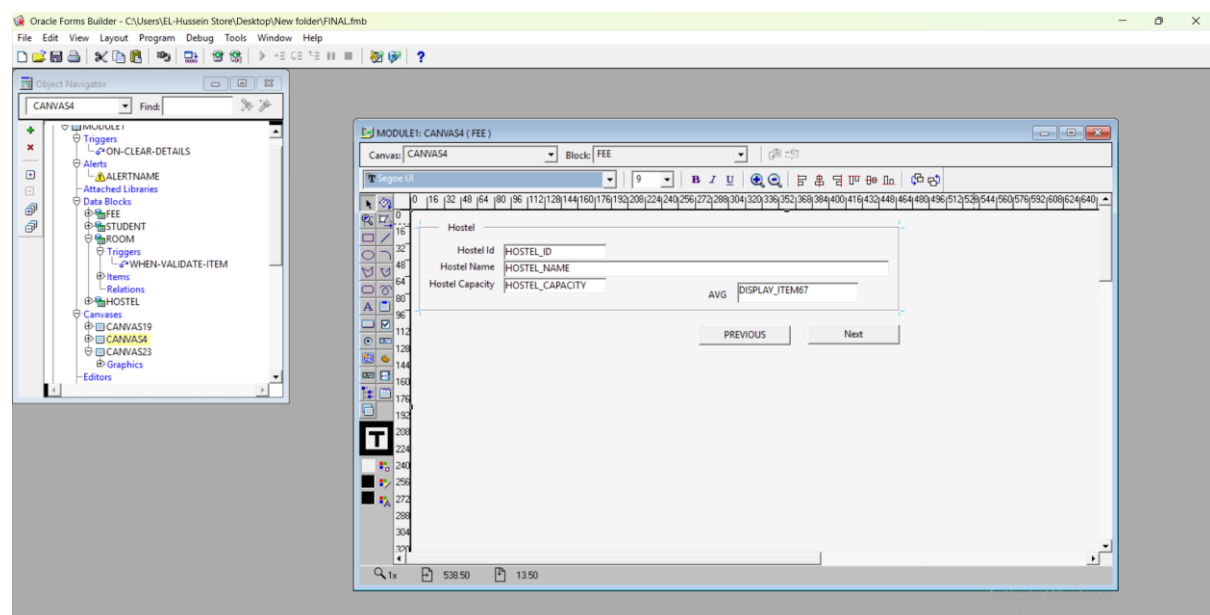
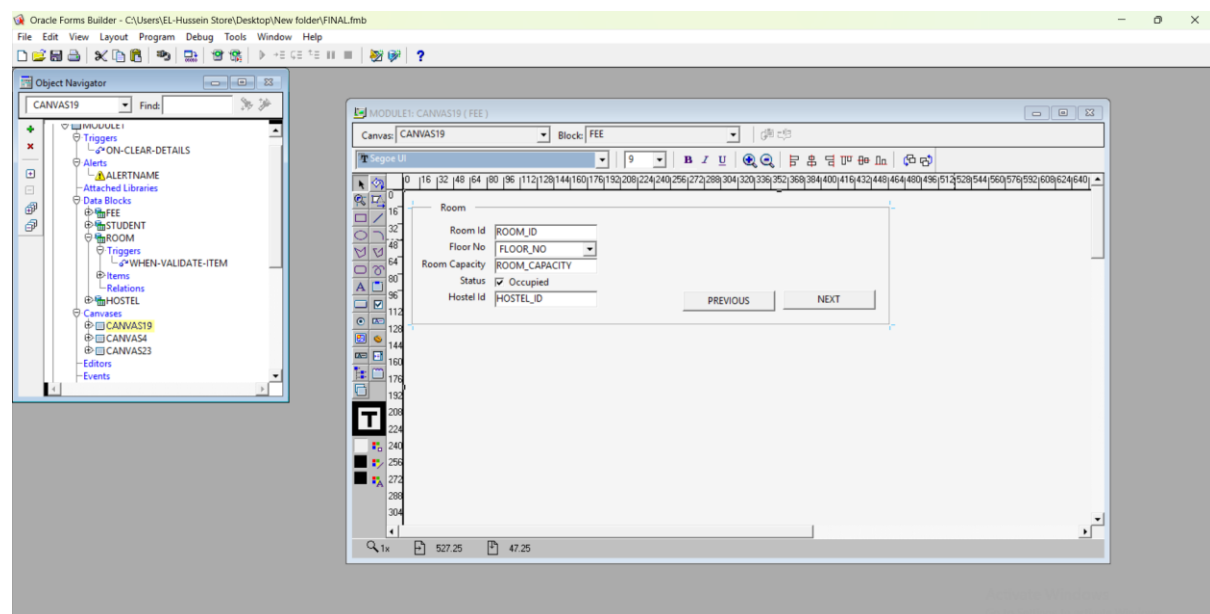
-- FEE

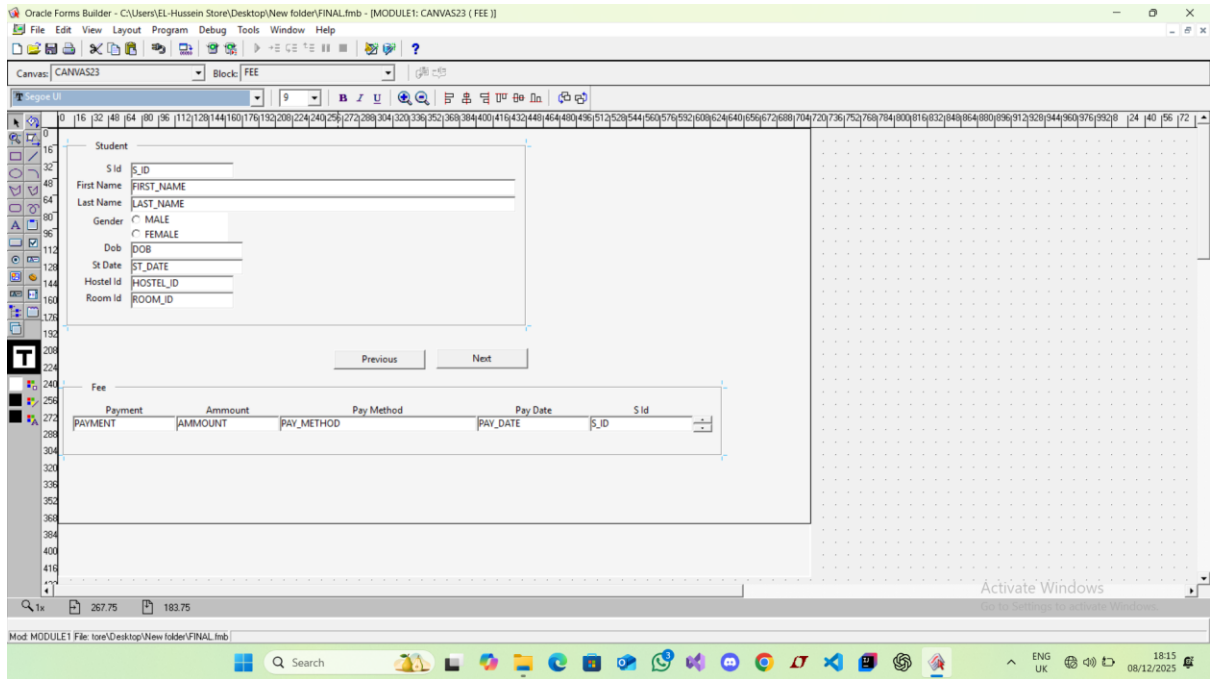
INSERT INTO FEE VALUES (1, 1500, 'Credit Card', DATE '2024-10-01', 1001);

INSERT INTO FEE VALUES (2, 1500, 'Cash', DATE '2024-10-02', 1002);

INSERT INTO FEE VALUES (3, 1500, 'Bank Transfer', DATE '2024-10-03', 1003);

Canvases





Check Box

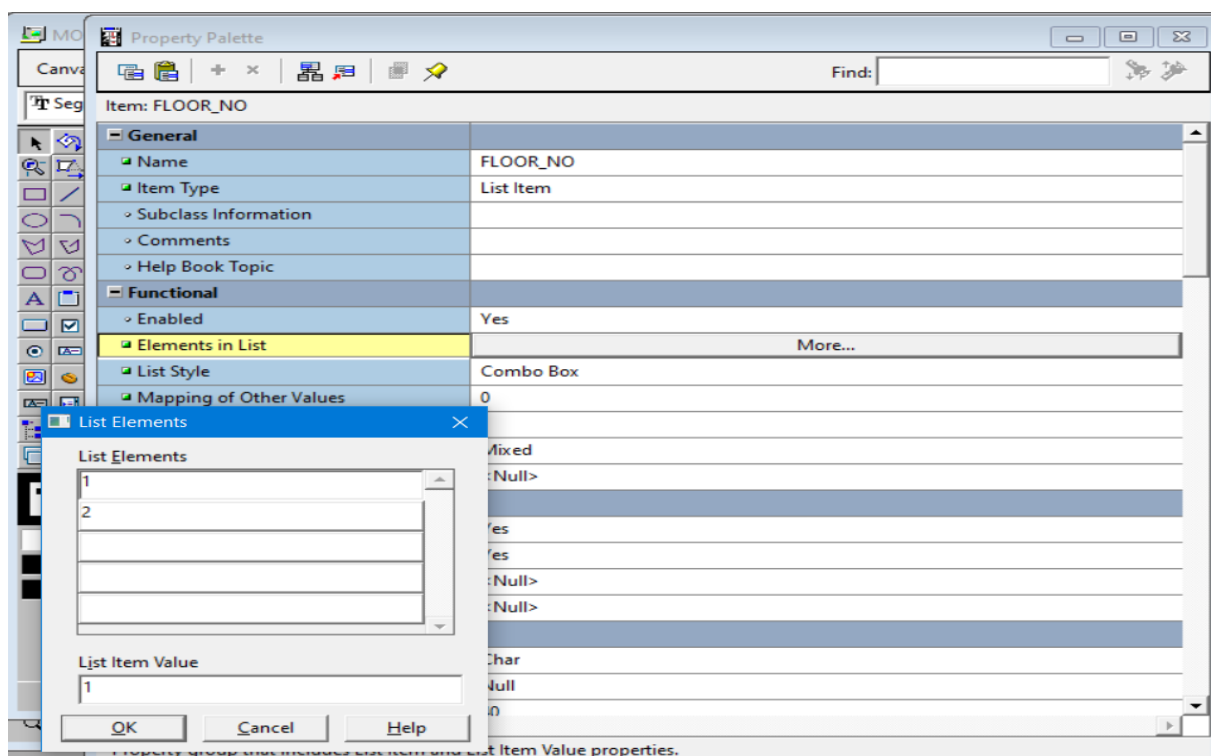
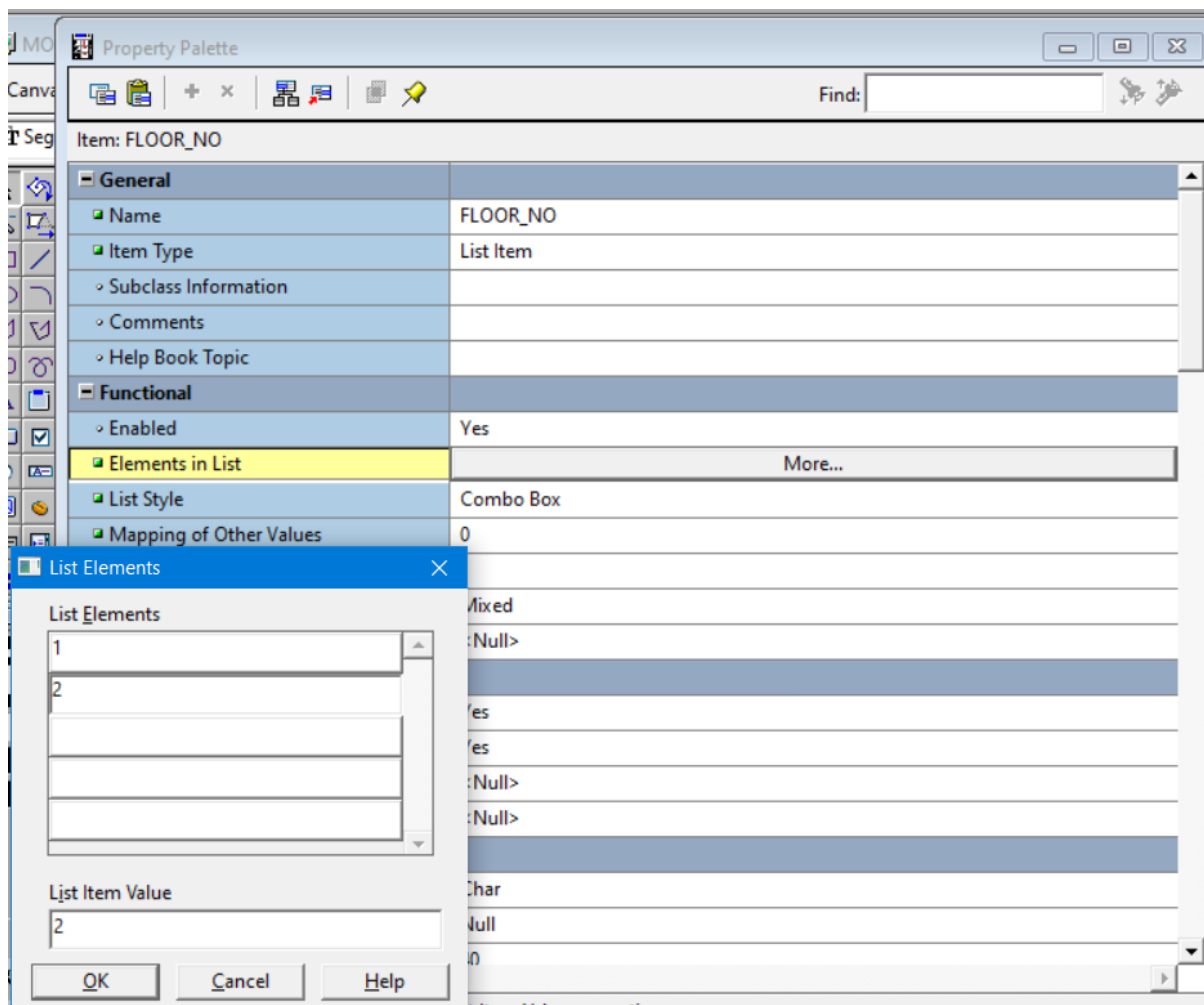
Item: STATUS	
General	
Name	STATUS
Item Type	Check Box
Subclass Information	
Comments	
Help Book Topic	
Functional	
Enabled	Yes
Label	Occupied
Access Key	
Implementation Class	
Value when Checked	Occupied
Value when Unchecked	Vacant
Check Box Mapping of Other Values	Checked
Popup Menu	<Null>
Navigation	
Keyboard Navigable	Yes
Mouse Navigate	Yes
Previous Navigation Item	<Null>
Next Navigation Item	<Null>
Data	
Data Type	Char
Data Length Semantics	BYTE
Maximum Length	40
Initial Value	
Copy Value from Item	
Synchronize with Item	<Null>
Calculation	
Calculation Mode	None
Formula	
General	

Radio Buttons

Radio Button: FEMALE	
General	
Name	FEMALE
Subclass Information	
Comments	
Functional	
Enabled	Yes
Label	FEMALE
Access Key	
Radio Button Value	F
Records	
Distance Between Records	126
Physical	
Visible	Yes
X Position	70
Y Position	95
Width	93
Height	14
Visual Attributes	
Visual Attribute Group	DEFAULT
Prompt Visual Attribute Group	DEFAULT
Color	
Foreground Color	black
Background Color	white
Fill Pattern	<Unspecified>
Font	
Font Name	Segoe UI
Font Size	9
Font Weight	Demilight
Font Style	Plain
Font Spacing	Normal
General	

Radio Button: MALE	
General	
Name	MALE
Subclass Information	
Comments	
Functional	
Enabled	Yes
Label	MALE
Access Key	
Radio Button Value	M
Records	
Distance Between Records	126
Physical	
Visible	Yes
X Position	70
Y Position	81
Width	93
Height	14
Visual Attributes	
Visual Attribute Group	DEFAULT
Prompt Visual Attribute Group	DEFAULT
Color	
Foreground Color	black
Background Color	white
Fill Pattern	<Unspecified>
Font	
Font Name	Segoe UI
Font Size	9
Font Weight	Demilight
Font Style	Plain
Font Spacing	Normal
General	

Combo Box



Item: FLOOR_NO

General	
Name	FLOOR_NO
Item Type	List Item
Subclass Information	
Comments	
Help Book Topic	
Functional	
Enabled	Yes
Elements in List	
List Style	Combo Box
Mapping of Other Values	0
Implementation Class	
Case Restriction	Mixed
Popup Menu	<Null>
Navigation	
Keyboard Navigable	Yes
Mouse Navigate	Yes
Previous Navigation Item	<Null>
Next Navigation Item	<Null>
Data	
Data Type	Char
Data Length Semantics	Null
Maximum Length	40
Initial Value	1
Required	Yes
Copy Value from Item	
Synchronize with Item	<Null>
Calculation	
Calculation Mode	None
Formula	

General

List of Values (LOV)

LOV Wizard

Record Groups can be based on SQL queries. Do you want to enter or modify the query that your LOV's Record Group uses?

If so, you may use the Query Builder by clicking Build SQL Query. Or, you may enter your query directly into the SQL Query Statement field below.

SQL Query Statement

```
SELECT ALL STUDENT.S_ID, STUDENT.FIRST_NAME, STUDENT.LAST_NAME,
STUDENT.GENDER,
STUDENT.DOB, STUDENT.ST_DATE, STUDENT.HOSTEL_ID,
STUDENT.ROOM_ID
FROM STUDENT
```

☒ Automatically size columns

Finish

General	
Name	LOV98
Subclass Information	
Comments	
Functional	
Title	STUDENT
List Type	Record Group
Record Group	LOV98
Column Mapping Properties	
Filter Before Display	No
Automatic Display	Yes
Automatic Refresh	Yes
Automatic Select	No
Automatic Skip	No
Automatic Position	Yes
Automatic Column Width	Yes
Physical	
X Position	0
Y Position	0
Width	360
Height	158
Visual Attributes	
Visual Attribute Group	DEFAULT
Color	
Foreground Color	<Unspecified>
Background Color	<Unspecified>
Fill Pattern	<Unspecified>
Font	
Font Name	<Unspecified>
Font Size	<Unspecified>

Display LOV automatically on navigation into item to which LOV is attached?

Trigger

Property Palette

Find:

Item: NEXT

General	
Name	NEXT
Item Type	Push Button
Subclass Information	
Comments	
Help Book Topic	
Functional	
Enabled	Yes
Label	NEXT
Access Key	
Implementation Class	
Iconic	No
Icon Filename	
Default Button	No
Popup Menu	<Null>
Navigation	
Keyboard Navigable	Yes
Mouse Navigate	Yes
Previous Navigation Item	<Null>
Next Navigation Item	<Null>
Records	
Current Record Visual Attribute Group	<Null>
Distance Between Records	0

General

PL/SQL Editor

Name: WHEN-BUTTON-PRESSED

Type: Trigger Object: ROOM NEXT

```
1 next_record;
```

Not Modified Successfully Compiled

Next_record;

Property Palette

Find:

Item: PREVIOUS

General	
Name	PREVIOUS
Item Type	Push Button
Subclass Information	
Comments	
Help Book Topic	
Functional	
Enabled	Yes
Label	PREVIOUS
Access Key	
Implementation Class	
Iconic	No
Icon Filename	
Default Button	No
Popup Menu	<Null>
Navigation	
Keyboard Navigable	Yes
Mouse Navigate	Yes
Previous Navigation Item	<Null>
Next Navigation Item	<Null>
Records	
Current Record Visual Attribute Group	<Null>
Distance Between Records	0

General

PL/SQL Editor

Name: WHEN-BUTTON-PRESSED

Type: Trigger Object: ROOM PREVIOUS

```

1 previous_record;

```

Not Modified Successfully Compiled

Previous_record;

Alert

Alert: ALERTNAME	
General	
Name	ALERTNAME
Subclass Information	
Comments	
Functional	
Title	ALERT
Message	Capacity is Exceeded
Alert Style	Stop
Button 1 Label	OK
Button 2 Label	Cancel
Button 3 Label	
Default Alert Button	Button 1
Visual Attributes	
Visual Attribute Group	DEFAULT
Color	
Foreground Color	<Unspecified>
Background Color	<Unspecified>
Fill Pattern	<Unspecified>
Font	
Font Name	<Unspecified>
Font Size	<Unspecified>
Font Weight	<Unspecified>
Font Style	<Unspecified>
Font Spacing	<Unspecified>
International	
Direction	Default

PL/SQL Editor

Name: WHEN-VALIDATE-ITEM

Type: Trigger Object: ROOM (Data Block Level)

```

1 Declare
2 b number;
3 Begin
4 If :ROOM.ROOM_CAPACITY > 5 then
5 b:= show_alert ('alertname');
6 If b= 88 then
7 Message ('you pressed the first butto');
8 Elsif b= 89 then
9 Message ('you pressed the second button ');
10 End if;
11 End if;
12 End;

```

Not Modified Successfully Compiled

```

Declare
b number;
Begin
If :ROOM.ROOM_CAPACITY > 5 then
b:= show_alert ('alertname');
If b= 88 then
Message ('you pressed the first button');
Elsif b= 89 then
Message ('you pressed the second button ');
End if;
End if;
End;

```

Master Detail

Relation: STUDENT_FEE	
General	
Name	STUDENT_FEE
Relation Type	Join
Subclass Information	
Comments	
Functional	
Detail Data Block	FEE
Join Condition	FEE.S_ID = STUDENT.S_ID
Delete Record Behavior	Non Isolated
Prevent Masterless Operations	No
Coordination	
Deferred	No
Automatic Query	No



You may optionally create and delete master-detail relationships to other data blocks in your form.

Create Relationship...

☒ Auto-join data blocks

Delete Relationship

Detail Item

PAYMENT

Master Data Blocks

STUDENT

Master Item

(No Join)

Join Condition

FEE1.S_ID = STUDENT.S_ID

Cancel

Help

< Back

Next >

Finish

Calculated Fields (Summary)

Item: DISPLAY_ITEM67	
Data	
• Data Type	Number
• Data Length Semantics	Null
• Maximum Length	30
• Initial Value	
• Format Mask	
• Copy Value from Item	
• Synchronize with Item	<Null>
Calculation	
• Calculation Mode	Summary
• Formula	
• Summary Function	Avg
• Summarized Block	HOSTEL
• Summarized Item	HOSTEL_CAPACITY
Records	
• Current Record Visual Attribute Group	<Null>
• Distance Between Records	0
• Number of Items Displayed	0
Database	
• Database Item	No
• Column Name	
• Primary Key	No
• Query Only	No
Physical	
• Visible	Yes
• Canvas	CANVAS4
• Tab Page	<Null>
• X Position	313
• Y Position	70
• Width	116
General	